

CHEMISTRY

Coupled conversion of polyethylene and carbon dioxide catalyzed by a zeolite–metal oxide system

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Zeolite-catalyzed polyethylene (PE) aromatization achieves reduction of the aromatic yield via hydrogenation and hydrogenolysis reactions. The hydrogen required for CO₂ hydrogenation can be provided by H radicals formed during aromatization. In this study, we efficiently convert PE and CO₂ into aromatics and CO using a zeolite–metal oxide catalyst (HZSM-5 + CuZnZrO_x) at 380°C and under hydrogen- and solvent-free reaction conditions. Hydrogen, derived from the aromatization of PE over HZSM-5, diffuses through the Brønsted acidic sites of the zeolite to the adjacent CuZnZrO_x, where it is captured in situ by CO₂ to produce bicarbonate and further hydrogenated to CO. This favors aromatization while inhibiting hydrogenation and secondary hydrogenolysis reactions. An aromatic yield of 62.5 wt % is achieved, of which 60% consisted of benzene, toluene, and xylene (BTX). The conversion of CO₂ reaches values as high as 0.55 mmol g_{PE}^{−1}. This aromatization–hydrogen capture pathway provides a feasible scheme for the comprehensive utilization of waste plastics and CO₂.

INTRODUCTION

Petroleum-based plastics, with their massive-scale manufacture [368 million tonnes Mt in 2019], short circulation lifetime (<10 years), and difficult natural decomposition are responsible for plastic waste accumulation and severe environmental challenges (1–6). Over 258 Mt of waste plastics are now generated worldwide per year, with polyethylene (PE) being the top contributor (2, 7). Notably, waste PE is a large reservoir of C/H. In the past, postconsumer plastics were mainly treated by incineration (for energy recovery) and physical-mechanical (for recycling). However, both routes have large carbon footprints and substantial issues (8–10).

The chemical upcycling of plastic wastes into liquid fuels and high-value chemicals is highly attractive. However, the high dissociation energy (~372 kJ mol^{−1}) and random nature of C–C bond cleavage in PE main chains make PE resistant to selective chemical transformations. At high temperatures (>450°C), PE macromolecules are cleaved via a free radical pathway, in which random C–C bonds undergo cleavage to form gases, liquids, and waxes over a wide range of carbon length (C₁ to C₄₀) (11–13). Controlled catalytic hydrogenolysis at lower temperatures (<300°C) can further optimize product distribution (14–16) at the expense of generating low-value alkanes and requiring large amounts of H₂ as a co-reactant.

Aromatic hydrocarbons (monocyclic, C_nH_{2n-6}, *n* ≥ 6) are intermediates for fuel additives, synthetic polymers, and surfactants that have been traditionally manufactured through naphtha reforming. The conversion of PE waste-derived C/H resources into aromatics is a more appropriate target product (17–20). ZSM-5 zeolites, with a unique three-dimensional MFI topology and a proper number of Brønsted acid active sites, can depolymerize PEs into aromatics by catalyzing C–C bond cleavage, aromatization, and hydrogen transfer processes (21, 22). However, the product distribution of depolymerization shows a strong temperature

dependence (23). PE depolymerization over HZSM-5 zeolite tends to produce C₁ to C₅ gases (>50%) under temperature conditions of 450° to 650°C, resulting in low aromatic yields (<40%) (22, 24). In addition, the aromatization of hydrogen-rich PE (i.e., effective H/C = 2.0) involves hydrogen transfer reactions and hydrogen evolution reactions (Fig. 1, route 1). These reactions induce secondary reactions such as multiple hydrocracking of long-chain alkanes to small gaseous alkanes and hydrogenation of aromatic precursor olefins (Fig. 1, route 2).

CO₂, as a waste carbon resource, requires a large amount of hydrogen to facilitate the conversion (25–27). Gao *et al.* (28) and Ni *et al.* (29) developed bifunctional zeolite–metal oxide (ZEO-OX) catalytic systems able to convert CO₂ into olefins and aromatics under high-pressure conditions at 320° to 400°C and H₂/CO₂ = 3. The active sites and working temperature of catalytic PE depolymerization and CO₂ hydrogenation are similar, thereby allowing effective coupling of PE depolymerization with CO₂ conversion and avoiding the use of large amounts of H₂. The co-conversion of waste carbon resources from plastics and CO₂ into high-value fuels and chemicals (30, 31) can provide a favorable solution for achieving global carbon neutrality targets.

In this work, we proposed an intensified process involving the coupling of aromatization and hydrogen capture using a uniquely designed low-cost ZEO-OX catalyst. We in situ captured H species generated from PE aromatization using a CO₂ hydrogen acceptor to promote the aromatization reaction. Moreover, secondary hydrogenation and hydrocracking reactions of aromatic precursor olefins were suppressed, achieving high-efficiency conversion of CO₂ and PE to aromatics and CO. A 62.5 wt % yield of aromatics was obtained using a physically mixed HZSM-5 + CuZnZrO_x catalyst at a moderate temperature of 380°C under hydrogen- and solvent-free reaction conditions. The ZEO-OX catalyst system exhibited good recycling and regeneration properties.

RESULTS AND DISCUSSION

ZEO-OX catalyst system construction

First, we used the one-pot method to convert low-density polyethylene [LDPE; molecular weight (*M_w*) ~1.46 × 10⁵ g mol^{−1}, dispersity

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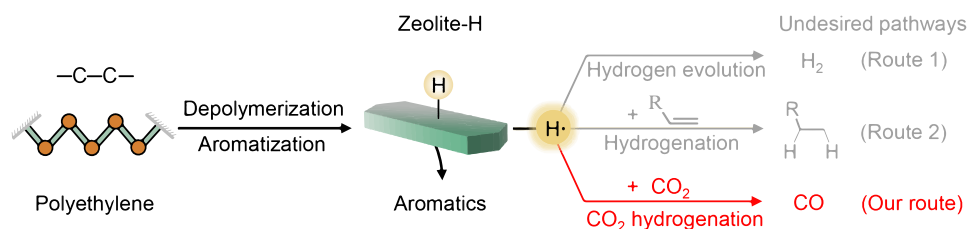


Fig. 1. Hydrogen atom transfer pathway of H species on the zeolite surface in the aromatization reaction.

(D) = 3.40; fig. S1] and CO_2 (5 bar). The experiments were conducted on a small high-temperature stainless steel reactor (fig. S2). In a blank test at 380°C for 200 min, liquid saturated alkanes were the main products (major chain lengths C_5 to C_{27} , yield: 67.8 wt %) and solid waxes (yield: 27.7 wt %), with virtually undetectable formation of aromatics (yield: <3.0 wt %). Subsequently, we selected several zeolites (HMOR, HY, HBeta, and HZSM-5, with Si/Al ratios of 25 to 30) known for their C—C bond cleavage ability to perform catalytic depolymerization experiments. LDPE (3.0 g) was combined with zeolite (0.2 g), and the reaction was conducted at 380°C under 5 bar of CO_2 . The resulting gases and both soluble and insoluble solids were collected, and CHCl_3 was used as the solvent. Liquid (soluble) products were analyzed by coupled to gas chromatography coupled with mass spectrometry (GC-MS; tables S1 and S2) and gas chromatography with a flame ionization detector (figs. S3 and S4) and quantified by mass balance (tables S3 to S5). The gases and solids were quantified via gas chromatography (fig. S5 and table S6) and thermogravimetric analysis (fig. S6), respectively. The product yields over various zeolites are provided in Table 1 (entries 2 to 6). Three zeolites, namely, HMOR, HY, and HBeta, cracked completely LDPE although the yield of aromatics was low (<20 wt %). The spent zeolites turned brown or black after the reaction, and O_2 -TG analysis confirmed the presence of numerous coke species (fig. S7, B to D), especially for the HY zeolites, which exhibited a weight loss of 41.6%. These results are consistent with previous observations detailing coke formation on zeolite catalysts. In addition, no CO_2 consumption was detected in both the blank experiment (Table 1, entry 1) and zeolite-catalyzed LDPE depolymerization experiments (Table 1, entries 2 to 4). (For the definition of CO_2 consumption, see the “Product analysis” section in the Supplementary Materials.)

The catalytic conversion of LDPE over HZSM-5(25) with thin nanosheets showed no macromolecular fragments, as revealed by gel permeation chromatography analysis (fig. S8). This suggests complete conversion of LDPE, yielding aromatics at 46.8 wt % and benzene-toluene-xylene (BTX) yield at 20.8 wt %. Upon reduction of the acid content of the zeolite, the yield of aromatics on HZSM-5 (Si/Al = 160) increased to ~50.7 wt %. Moreover, it showed an increase in the yields of gas and solid when the Si/Al ratio was reduced from 160 to 25, i.e., the gas yield increased from 26.8 to 28.6 wt %, and the coke yield increased from 0.3 to 0.8 wt % (table S7). Abundant acidic sites (low Si/Al ratio) may cause excessive coupling reaction of aryl groups through a carbonium ion reaction mechanism. Mild Brønsted acidity assists in the inhibition of β -scission and coke formation (32, 33). In addition to its remarkable activity and product selectivity, HZSM-5(160) demonstrated excellent resistance to coke deposition, unlike other zeolites (HMOR, HBeta, and HY). This was evident from the light color of the HZSM-5(160) catalyst after the reaction and the coke signal in the

O_2 -TG analysis results (weight loss: 6.9%; figs. S7A and S9). The special MFI type topology and channels provide high aromatic yields and excellent coking resistance (34, 35) in contrast to HMOR, HBeta, and HY (table S7). CO_2 consumption was achieved over HZSM-5(25) and HZSM-5(160) (0.15 and 0.11 $\text{mmol g}_{\text{PE}}^{-1}$, respectively; Table 1, entries 5 and 6). This result suggests that active hydrogen species adsorbed on the HZSM-5 zeolite surface [zeolite-H (Z-H)] during the formation of aromatics were involved in the CO_2 reduction. The single-component acidic HZSM-5 zeolite features low activity in converting acidic CO_2 molecules. However, CO_2 consumption was not detected on HMOR, HY, and HBeta zeolites, which was attributed to limited apparent hydrogen release (<0.3 $\text{mmol g}_{\text{PE}}^{-1}$; fig. S10) as a result of low aromatic yields (<20 wt %). The more amount of active hydrogen species is generated in companion to the produced aromatics, which is a prerequisite for CO_2 conversion. Besides, the HMOR, HY, and HBeta zeolite surfaces were covered with a large amount of coke, which may hinder CO_2 adsorption (table S7).

Subsequently, transition metal catalysts (Cu, Fe, Ni, and Co; figs. S11 and S12) with CO_2 activation capability were physically mixed with HZSM-5 (Si/Al = 160). Mixture of HZSM-5 with transition metal composite oxides [composed of MZnZrO_x , M: Zn:Zr (molar ratio) = 5:2:2; mass ratio 2:1] resulted in notably higher CO_2 conversion efficiencies. Thus, CO_2 consumption over HZSM-5 mixed with CoZnZrO_x , NiZnZrO_x , FeZnZrO_x , and CuZnZrO_x reached 0.28, 0.35, 0.48, and 0.55 $\text{mmol g}_{\text{PE}}^{-1}$, respectively. The CO_2 consumption value on CuZnZrO_x was approximately five times as that achieved over the single-component HZSM-5 zeolite. The combination of transition metal oxides CuZnZrO_x and HZSM-5 provided efficient CO_2 conversion and notably improved the yield of aromatics from 50.7 wt % (over HZSM-5) to 62.5 wt % (over HZSM-5 + CuZnZrO_x). However, the yield of aromatics over the HZSM-5 + CoZnZrO_x catalyst was reduced to 45.6 wt % owing to the strong dehydrogenation ability of Co (36, 37), which led to notable coke deposition during the excessive dehydrogenation of the reaction system (coke amount: 10.8 wt %; fig. S13 and table S8). Moreover, the catalyzed excessive cracking reaction generated large amount of gases (33.0 wt %, mainly CH_4 ; Table 1, entry 7, and table S8). Lower CO_2 consumption and aromatic yield were observed on NiZnZrO_x + HZSM-5, since NiZnZrO_x exhibits high hydrogenolysis activity and then promotes the conversion of aromatic precursors into substantial quantities of CH_4 and C_2H_6 (table S8). Thus, the active hydrogen is largely consumed, resulting in low efficiency in CO_2 consumption and aromatic yield. Compared to CuZnZrO_x , both yields of aromatics and BTX were reduced on FeZnZrO_x mixed with HZSM-5, although they have a similar CO_2 consumption (Table 1, entry 9). This possible reason is that FeZnZrO_x promotes the coupling reaction of monocyclic

Table 1. Catalytic performance of several catalysts for the simultaneous PE and CO₂ conversion. Reaction conditions: Low-density polyethylene (LDPE, $M_w = 1.46 \times 10^5 \text{ g mol}^{-1}$, $\bar{D} = 3.40$) 3.0 g, zeolite 0.2 g, MZnZrO_x (molar ratio M:Zn:Zr = 5:2:2) 0.1 g, 5 bar CO₂, 200 min.

Entry	Catalyst	T (°C)	Y_{Gas} (wt %)	Y_{Liquid} (wt %)	Y_{Solid} (wt %)	Aromatics yield* (wt %)	BTX yield* (wt %)	CO ₂ consumption† (mmol g _{PE} ⁻¹)
1	None	380	4.5	67.8	27.7	< 3.0	Not detected	Not detected
2	HMOR(25)	380	8.3	90.2	1.5	6.8	<1.0	Not detected
3	HY(30)	380	8.8	85.9	5.3	10.3	1.5	Not detected
4	HBeta(25)	380	14.1	79.6	6.3	18.2	3.7	Not detected
5	HZSM-5(25)	380	28.6	70.3	1.1	46.8	20.8	0.15
6	HZSM-5(160)	380	26.8	72.6	0.6	50.7	27.2	0.11
7‡	HZSM-5 + CoZnZrO _x	380	33.0	56.0	11.0	45.6	31.4	0.28
8‡	HZSM-5 + NiZnZrO _x	380	28.5	69.3	2.2	57.2	37.3	0.35
9‡	HZSM-5 + FeZnZrO _x	380	23.9	75.6	0.5	57.1	31.5	0.48
10‡	HZSM-5 + CuZnZrO _x	380	24.4	75.4	0.2	62.5	37.2	0.55
11‡	HZSM-5 + CuZnZrO _x	340	16.9	82.3	0.9	50.1	23.5	0.21
12‡	HZSM-5 + CuZnZrO _x	360	21.5	77.6	0.9	58.0	29.8	0.29
13‡	HZSM-5 + CuZnZrO _x	400	29.1	70.2	0.7	58.5	34.2	0.57
14§	Catalyst recycle 1	380	26.7	71.3	2.0	58.9	35.8	0.57
15§	Catalyst recycle 2	380	25.4	71.5	3.1	55.9	33.5	0.52
16§	Catalyst recycle 3	380	19.6	76.8	3.6	52.8	29.7	0.49
17§	Catalyst regeneration	380	26.8	72.5	0.7	60.1	36.3	0.57

*The yields of aromatics and benzene, toluene, and xylene (BTX) were based on the weight ratio of the raw LDPE. †The CO₂ consumption was defined as the moles of detected CO divided by the mass of the raw LDPE. The carbon balance of CO₂ closed to values greater than 95%. ‡In entries 7 to 13, HZSM-5 with a Si/Al ratio of 160 was adopted. §Catalyst and reaction conditions for entry 14 were used for catalyst recycling and catalyst regeneration experiments.

aromatics, leading to the covered acid active site by the formed coke. The above results indicate that a moderate hydrogen transfer and hydrogenation capacity of the metal can be ensured through the combination of MZnZrO_x with HZSM-5. Notably, the spent HZSM-5 + CuZnZrO_x catalysts showed stable performance for at least three cycle runs without requiring any additional treatment. It also showed good regeneration performance (Table 1, entries 14 to 17, and fig. S14).

Characterization of catalytic systems and expansion of PE raw materials

The structures of the HZSM-5, CuZnZrO_x , and HZSM-5 + CuZnZrO_x catalysts were characterized by a number of techniques. According to scanning electron microscopy (SEM) and transmission electron microscopy (TEM) analyses, HZSM-5 exhibited an *a*-axis orientation length of $\sim 3 \mu\text{m}$ and a stereo hexagonal structure (figs. S16 and S17A). X-ray diffraction (XRD) analysis showed good crystallinity (fig. S12), while N₂ adsorption analysis revealed a high surface area ($452.7 \text{ m}^2 \text{ g}^{-1}$; fig. S18). CuZnZrO_x is composed of $\sim 30\text{-nm}$ crystallites that exist as agglomerated particles of nanometric size. The surface area was $62.1 \text{ m}^2 \text{ g}^{-1}$, with a uniform mesoporous pore size of 3.5 nm (figs. S12, S17B, and S19). Crystallinity was well maintained after the physical mixing of HZSM-5 and CuZnZrO_x (fig. S12), and energy dispersive x-ray

(EDX) elemental mapping showed good contacts on the structural domains of the catalyst (figs. S20 and S21). Thus, according to the preliminary experimental results and characterizations, the acid sites on HZSM-5 were responsible for PE C—C bond breaking and aromatization reactions, while CuZnZrO_x with metal sites catalyzed the CO₂-participating hydrogenation reduction reaction (Fig. 2A).

Nuclear magnetic resonance (NMR) results of the liquid products are shown in Fig. 2B. The ¹³C spectrum featured signals in the aromatic [120 to 150 parts per million (ppm)] and aliphatic hydrocarbon (10 to 30 ppm) regions, most of which corresponded to the cyclic carbons of the aromatic ring. The ¹H spectrum revealed most of the protons to bond to monocyclic aromatics (6.5 to 7.5 ppm) and alkyl groups on the monocyclic aromatics (2.0 to 3.0 ppm), whereas fused aromatic rings bound fewer protons (7.5 to 8.5 ppm) (38). The carbon number distribution of the detailed products showed a typical bimodal distribution with C₃ to C₄ aliphatic hydrocarbons and C₇ to C₈ aromatic hydrocarbons being the main products (Fig. 2C). Furthermore, we used the ZEO-OX catalytic system for the upgrading of HDPE (high-density polyethylene) and actual postconsumer plastics (including self-sealing bags, plastic films, and laundry detergent bottles; fig. S22) under a CO₂ atmosphere. Aromatic yields as high as 54.6 to 58.6 wt % were obtained under these materials, and the CO₂ conversion all exceeded 0.40 mmol

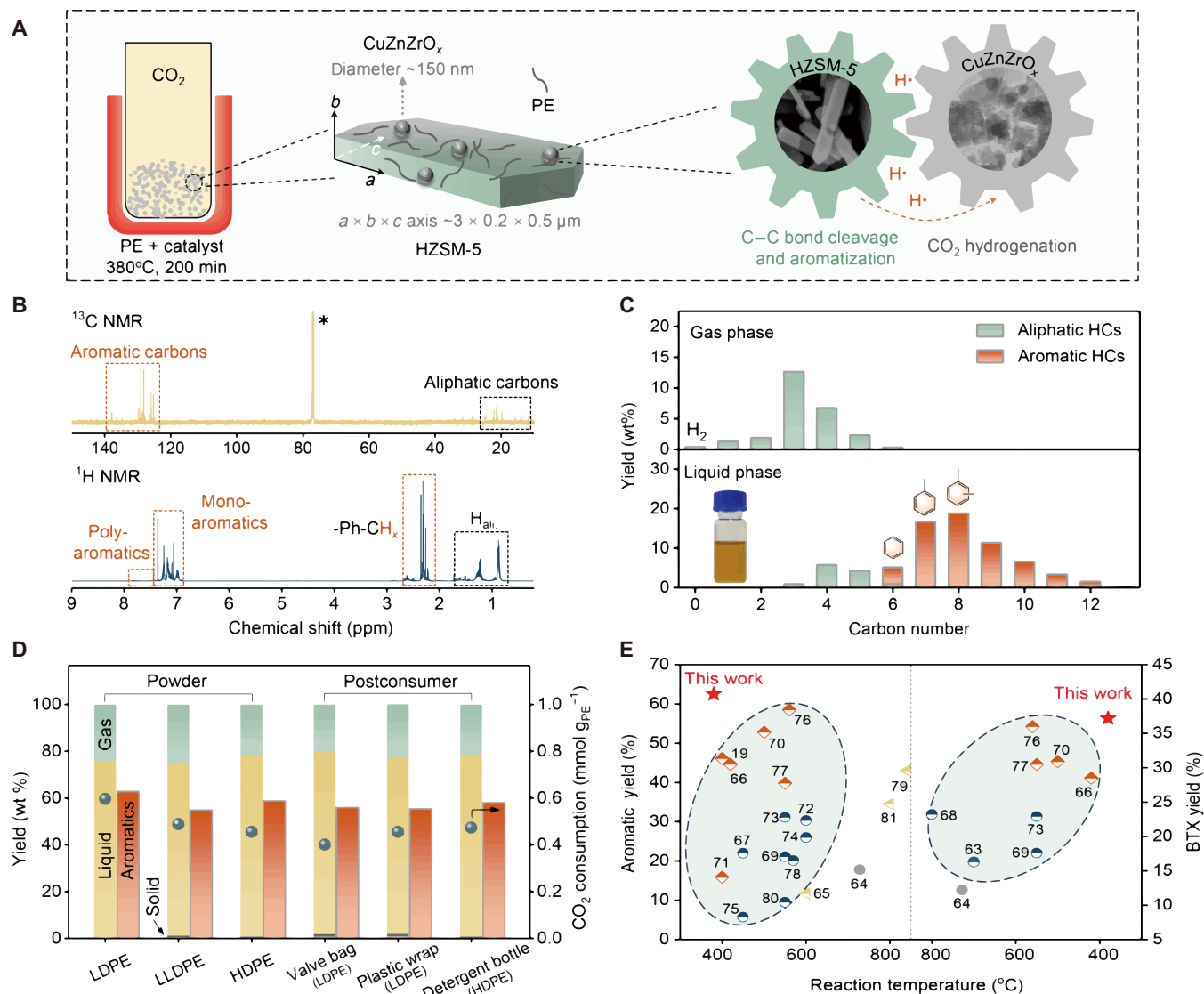


Fig. 2. Results from conversion of PE and CO₂ catalyzed by ZEO-OX. Experiments involving the conversion of PE and CO₂ were conducted over physically mixed HZSM-5 + CuZnZrO_x for 200 min at 380°C with 5 bar CO₂. (A) Reaction system and illustration of the structure for the HZSM-5 + CuZnZrO_x catalyst. (B) ¹³C and ¹H NMR spectra of liquid products from LDPE, analyzed in deuterated CDCl₃. (C) Distribution of the carbon number in products from LDPE conversion. HCs, hydrocarbons. (D) Product distributions for raw PE powder and postconsumer PE. (E) Comparison of aromatic and BTX yields from PE conversion. The yields of aromatics and BTX are shown in table S7.

g_{PE}⁻¹ (Fig. 2D and fig. S23). A comparison of the results with reported data reveals that generally metal-promoted zeolites (orange data points, Fig. 2E and table S13) are beneficial for the production of aromatics. In contrast, PE conversion over CO₂-containing ZEO-OX yielded high aromatic production (62.5 wt %; Fig. 2E) at a relatively low temperature of 380°C, along with a BTX yield (37.2 wt %) substantially higher than those reported in the literature.

Proximity-dependent aromatic generation and CO₂ hydrogen capture

We tested the HZSM-5 + CuZnZrO_x tandem catalysts in an upgraded PE + CO₂ scheme. Then, we explored the respective roles of HZSM-5

and CuZnZrO_x (figs. S24 to S26). When the single-component CuZnZrO_x was used to crack LDPE (fig. S27), similar yields of gas-liquid-solid products to those achieved via LDPE pyrolysis were obtained, with the major products being alkanes. Only a slight increase in apparent H₂ evolution occurred (0.15 mmol g_{PE}⁻¹; fig. S28), suggesting that the Lewis acid sites on CuZnZrO_x (2.15 μmol g_{PE}⁻¹; fig. S29) were unable to break C–C bonds (fig. S24). The pyrolysis of LDPE on HZSM-5 was not affected by the atmosphere, whether it was CO₂ or N₂ (fig. S30). HZSM-5 + CuZnZrO_x showed higher aromatic yields than HZSM-5 (Table 1, entry 6, and Fig. 3A) since the metal moiety of CuZnZrO_x accelerated the evolution of H species formed on HZSM-5 (1.30 mmol g_{PE}⁻¹ under a CO₂ atmosphere and 1.41 mmol g_{PE}⁻¹ under a N₂ atmosphere; fig. S28). In the

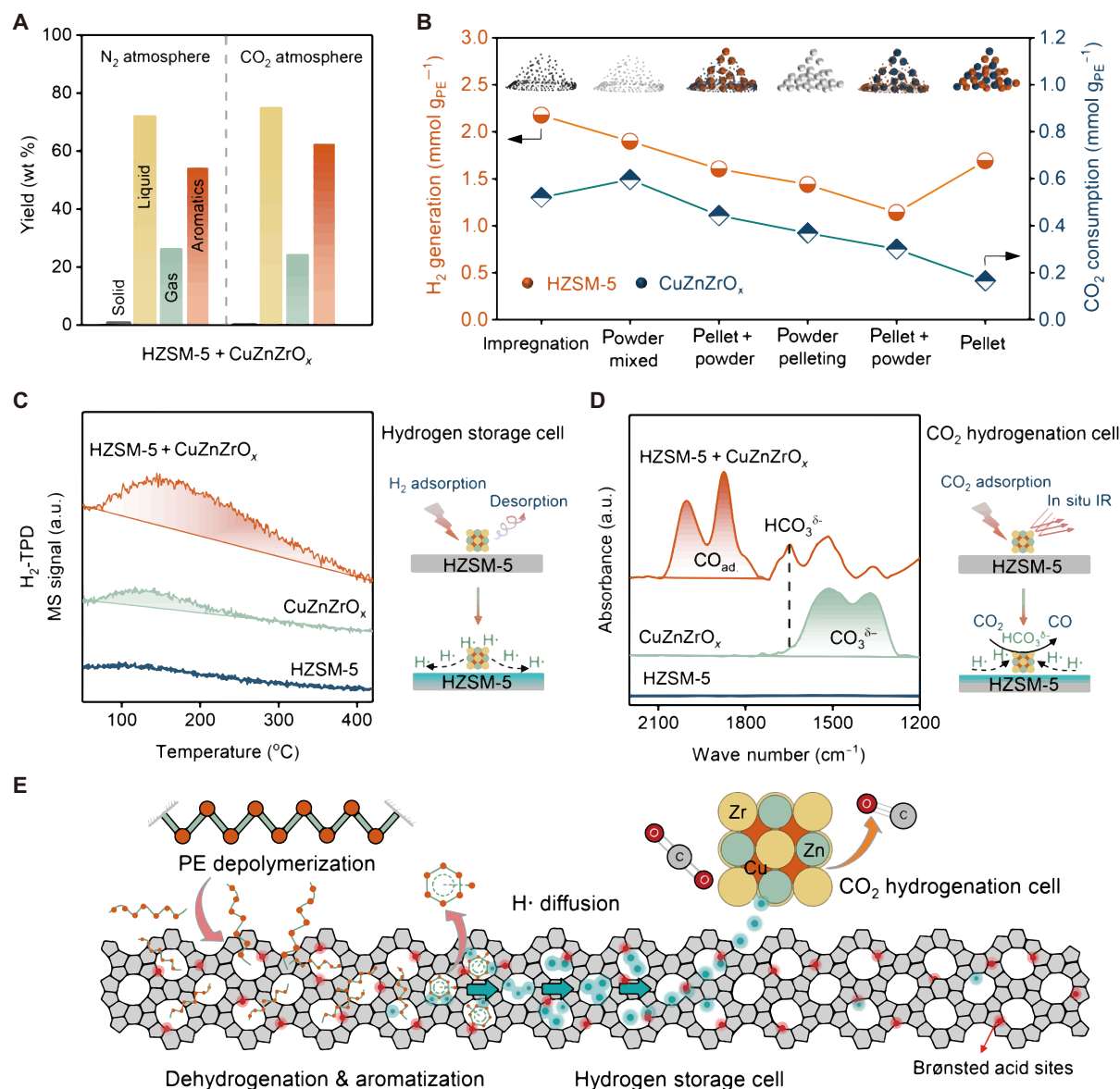


Fig. 3. Investigation of the roles of individual metal and acid sites for the co-conversion of PE and CO₂. (A) Effects of CO₂ and N₂ on PE conversion over HZSM-5 + CuZnZrO_x catalysts. (B) Reaction results over HZSM-5 and CuZnZrO_x catalysts packed in different configurations. Illustrations show the states of the different mixing configurations. (C) H₂-TPD profiles of HZSM-5, CuZnZrO_x, and HZSM-5 + CuZnZrO_x. (D) In situ diffuse reflectance Fourier transform infrared spectra of CO₂ adsorption and reaction on three catalysts treated with hydrogen at 230°C for 3 hours. The spectra were collected at 300°C. (E) Proposed hydrogen capture and CO₂ hydrogenation mechanism for PE-CO₂ co-conversion in the presence of Brønsted acid sites of HZSM-5 and CuZnZrO_x. a.u., arbitrary units.

dehydrogenative-aromatization reaction, the removal of H species was considered as the rate-limiting step in the generation of aromatics from olefinic intermediates. The presence of metal active sites improved the aromatic selectivity by accelerating the consumption of H species, thus promoting the dehydro-aromatization of olefinic intermediates (39, 40). Another key reason explaining the increased aromatic yield in our ZEO-OX scheme was the hydrogen species elimination route involving CO₂ (Fig. 1), which avoided the formation of undesired alkanes from the hydrogenation of some of the aromatic precursor olefins (suppression of route 2 in Fig. 1). From the viewpoint of H/C equilibrium, higher aromatic yields are indicative of the formation of more hydrogen-rich products (e.g., H₂), in line

with the experimental results (fig. S28 and Table 1, entries 1, 6, and 10). Moreover, a high CO₂ consumption (from 0.11 to 0.55 mmol g_{PE}⁻¹) was found to correspond with high aromatic yield (from 50.7 to 62.5 wt %).

The proximity of the acid-metal active sites has been reported to play a key role in determining the activity of bifunctional catalysts (41–43). We studied the catalytic behavior of catalysts with different degrees of integration (Fig. 3B). Under a mixed particle configuration (250 to 350 μm, mode VI), the aromatic yield (table S9) and CO₂ consumption (0.17 mmol g_{PE}⁻¹) were similar to those obtained by the single-component HZSM-5 (Table 1, entry 6). However, there was still a substantial evolution of H₂ (1.67 mmol g_{PE}⁻¹). This can be

explained by hydrogen species formed on the zeolite (Z-H) during aromatization being hardly diffused over long distances to reach CuZnZrO_x and be captured by CO₂. Here, routes 1 and 2 dominated (Fig. 1). By transitioning from pellet mixing to powder and pellet mixing (Fig. 3B, modes III and V) and further to the physical dispersion of CuZnZrO_x on HZSM-5 microcrystals (Fig. 3B and fig. S20, mode II), we gradually make shorter distances of the acid-metal proximity. CO₂ consumption and the yield of aromatics improved notably over this catalyst (Fig. 3B and table S9). However, nanoscale proximity, that is, the impregnation of metal oxides on the HZSM-5 surface (Fig. 3B, mode I), has been shown to reduce catalytic activity (44), (i) a weight loss of 7.3% resulted in more coke being formed (fig. S31 and table S9); (ii) the total aromatic yield was reduced by 4.9 wt %; and (iii) CO₂ consumption was decreased by 15% (0.60 versus 0.52 mmol g_{PE}⁻¹). Concerning on the impregnated CuZnZrO_x/HZSM-5 sample, the Fourier transform infrared adsorption spectra of pyridine and collidine indicate the decrease in both internal and total Brønsted acid site densities from 2.3 to 1.2 and 3.0 to 1.7 μmol g⁻¹, respectively (fig. S32 and table S10). In addition, the impregnation approach resulted in the blockage of some of the micropores of the HZSM-5 zeolite (table S11). These results indicate that the decreased accessible active sites would reduce the aromatic yields (45, 46). Therefore, the H species produced during the aromatization of PE on the zeolite must diffuse into the metal oxide active site with the proper acid-metal proximity, where it combines with CO₂ to produce CO. Metal oxides facilitated the removal of hydrogen species from zeolites rather than inducing the hydrogenation or hydrogenolysis of the aromatic precursor olefins.

We further proposed a possible microreaction mechanism for PE aromatization–CO₂ hydrogen capture. As revealed by H₂ temperature-programmed desorption (TPD) (Fig. 3C), standalone HZSM-5 could not store hydrogen (47). Independent CuZnZrO_x showed a slightly enhanced H₂ desorption peak. In contrast, HZSM-5 + CuZnZrO_x exhibited H₂ adsorption peak areas one order of magnitude higher. Considering the respective adsorption amounts of HZSM-5 and CuZnZrO_x, most of the adsorbed hydrogen in the HZSM-5 + CuZnZrO_x mixture was stored on the HZSM-5 surface adjacent to CuZnZrO_x. Hydrogen dissociated from the metal sites can be stored within the HZSM-5 zeolite via hydrogen spillover, as described elsewhere (47, 48). In addition, the proximity effect between Brønsted acid and metal sites was confirmed through a comparison of the temperature-programmed reduction of H₂ (H₂-TPR) signals of HZSM-5, NaZSM-5, and S-1 mixed with CuZnZrO_x (fig. S33). The hydrogen storage sites may originate from the Brønsted acid sites in the HZSM-5 zeolite (49). Reducing the Si/Al ratio in HZSM-5 increases the amount of acid sites (fig. S34); subsequently, the reduction peaks of Cu species shifted to lower temperatures (fig. S35), indicating that the Brønsted acid sites on HZSM-5 triggered the hydrogen spillover proximity effect. Thus, the Brønsted acid sites are acting as a “hydrogen storage cell.”

To clarify the CO₂ hydrogen capture mode and reaction pathway during PE aromatization, we monitored the ZEO-OX catalyst by CO₂-TPD and in situ Diffuse Reflectance Fourier Transform Infrared Spectroscopy. Independent HZSM-5 sample exhibited negligible CO₂ adsorption (Fig. 3D and fig. S36), whereas CuZnZrO_x sample showed elevated CO₂ adsorption (3.3 μmol g⁻¹) and the formation of carbonate (CO₃^{δ-}) adsorbed species. In the presence of the hydrogen storage cell HZSM-5, CuZnZrO_x showed a high CO₂ adsorption capacity of 13.3 μmol g⁻¹. In addition, infrared vibrations showed the carbonates’

peaks as weakened, and hydrogenated bicarbonate species (HCO₃^{δ-}) emerged (50). Moreover, stronger CO adsorption peaks at 2055 and 1860 cm⁻¹ were observed (51, 52). Combining these results with those obtained by H₂-TPD (Fig. 3C), we can draw two substantial conclusions. First, although HZSM-5 showed limited hydrogen activation and CO₂ adsorption capabilities, this material was able to store hydrogen. Second, the active hydrogen provided by HZSM-5 could hydrogenate the carbonate species that were activated through adsorption on the surface of CuZnZrO_x, generating CO as a result (see Fig. 1). Therefore, in the HZSM-5 + CuZnZrO_x two-site catalyst system, HZSM-5 served as a hydrogen storage cell, while CuZnZrO_x functioned as the “CO₂ hydrogenation cell” architecture. Our PE catalytic upgrading scheme (Fig. 3E) has a unique hydrogen transfer pathway, which is distinguish from those of classical reaction pathways such as hydrogen overflow (Fig. 1, route 1) and olefin hydrogenation (Fig. 1, route 2) (45, 53).

Total reaction pathway for the upgrading of PE and CO₂

We comprehensively examined the *M_w* and product compositions of the recovered PE during a time evolution experiment to demonstrate the process of PE conversion involving CO₂ participation. The experiment was conducted at 380°C and 5 bar CO₂ using HZSM-5 zeolite and HZSM-5 + CuZnZrO_x as catalysts. Within the initial 50 min under catalysis by HZSM-5 + CuZnZrO_x, the fraction of soluble liquid hydrocarbons increased notably, appearing as residual PE *M_w* decreased rapidly from 1.46 × 10⁵ to 0.24 × 10⁵ g mol⁻¹ (figs. S37 to S39). Over time, *D* increased initially from 3.39 to 3.61 and then decreased to 1.85 (table S12). HZSM-5 + CuZnZrO_x achieved lower PE depolymerization rates than HZSM-5 (as indicated by the reaction time needed to achieve the same phase fraction; figs. S38 and S39), probably because CuZnZrO_x hindered the diffusion of fragments to the surface and pores of HZSM-5. This was favorable for controlling the formation of small molecules because the amount of the depolymerized fragment C₃ to C₉, which could enter the straight channels (5.3 × 5.6 Å) of HZSM-5 zeolite, did not considerably increase as it did under HZSM-5 catalysis. This, in turn, inhibited the hydrogenation of aromatic precursors and favored the removal of hydrogen species via CO₂ capture. Consequently, the formation of by-product alkanes was inhibited (fig. S40 and table S4). The yield of aromatics increased following CO₂ conversion (Fig. 4, B and C). At longer reaction times, the yield of aromatics reached a plateau of 62.5 wt %. H₂ evolution (Fig. 4A) and CO₂ consumption revealed the same trend as the yield of aromatics. However, extending the reaction time did not result in higher CO₂ consumption, despite the occurrence of a higher H₂ fraction (~15%) than in the initial stage (fig. S41) but rather resulted in an increase in the hydrogenolysis product CH₄/C₂H₆ (fig. S40). The potential reasons for this phenomenon are as follows: (i) Tests involving the variation in the initial CO₂ pressure showed a linear increase in CO₂ conversion with the initial pressure and a linear decrease in hydrogen evolution (fig. S42). The CO₂ hydrogenation with gaseous H₂ as spilled from HZSM-5 zeolite was considerably more difficult than radical hydrogen at the HZSM-5/CuZnZrO_x interface. (ii) A low CO₂ fraction led to a reduced likelihood of effective collisions between CO₂ and H₂ molecules (fig. S41).

We also conducted isotope-tracing experiments to determine whether the detected CO originated solely from CO₂ or whether some CO was created through a reverse Boudouard reaction involving

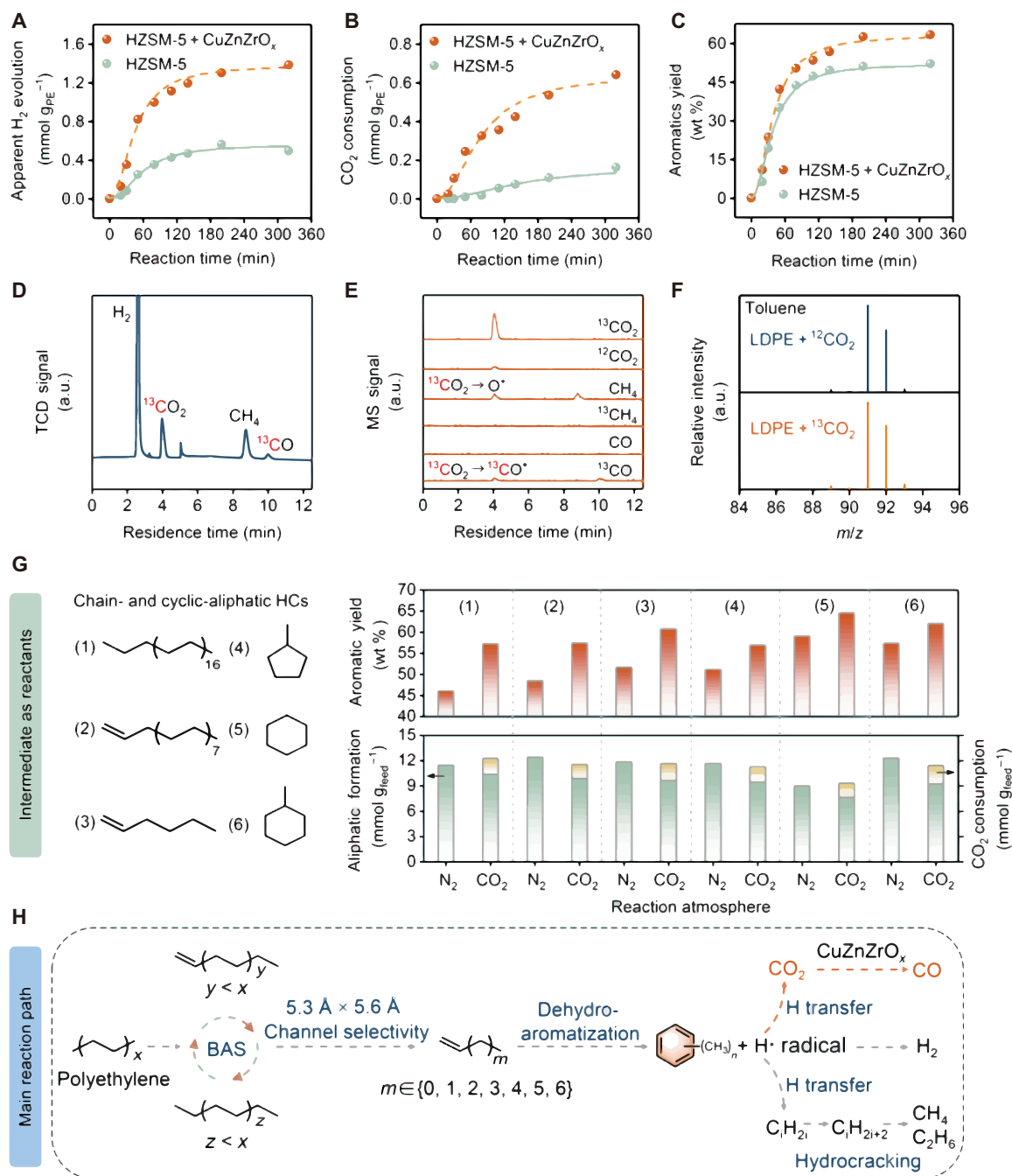


Fig. 4. Kinetics and mechanism of the catalytic upcycling of PE and CO_2 . Time course of the solvent-free conversion of PE and CO_2 . (A) Apparent hydrogen evolution; (B) CO_2 consumption; (C) aromatic yield. Product analysis for the ^{13}C isotope-labeled experiment. (D) Thermal conductivity detector (TCD) signals of the gas phase; (E) corresponding mass spectrometry (MS) spectra; (F) MS signals of aromatics (toluene). (G) Reaction performances of different model molecules including chain/cyclic aliphatic hydrocarbons over the ZEO-OX catalysts. Reaction conditions: 3.0 g hydrocarbon reactants, 20 bar CO_2 , 0.3 g HZSM-5 + CuZnZrO_x, 380°C, 200 min. (H) Proposed reaction pathway for the co-conversion of PE and CO_2 to aromatics and CO. m/z , mass/charge ratio. BAS, Brønsted acid site.

carbon from PE ($CO_2 + C = 2CO$). Experiments were performed using ^{13}C and unlabeled PE. Results revealed that all the CO originated from ^{13}C -labeled CO_2 (Fig. 4, D and E). This can be attributed to the high temperatures ($>700^\circ C$) required for the reverse Boudouard reaction (54, 55). Liquid product analysis showed that CO_2 was not incorporated into the aromatic rings and alkane chains (Fig. 4F and fig. S43).

To explore the reaction route of PE depolymerization, aromatization, and hydrogen capture, experiments were conducted under typical reaction conditions (Table 1, entry 10). Potential PE depolymerization fragments (i.e., long-chain saturated or unsaturated aliphatic hydrocarbons) and cyclized intermediate species (ring-forming saturated aliphatic hydrocarbons) were

used as reactants. They exhibited similar reactivity and bimodal product distribution compared with LDPE (Fig. 4G and fig. S44). Simultaneously, the utilization of CO₂ as a hydrogen acceptor notably increased aromatic yields and reduced the amount of undesired hydrogen transfer products such as aliphatic hydrocarbons. According to the product carbon distribution (Fig. 2C), depolymerized fragments (fig. S45), and validation of model reactants, the main reaction pathway for PE and CO₂ involved aromatization and hydrogen capture within the ZEO-OX catalytic system (Fig. 4H). Initially, molten PE formed large molecular long-chain olefins or alkanes via the β -scission of the polymer main chain C—C bonds at the Brønsted acid sites on the surface of HZSM-5 (11, 16, 56). Then, these preliminary depolymerized fragments continuously underwent successive β -scissions until they formed short-chain olefins that could diffuse through the 10-membered pore channels (5.3 Å $\times$ 5.6 Å) of HZSM-5 and contacted the acidic sites within the pores. Short-chain olefins (C₃ to C₉) were cyclized and dehydrogenated within the shape-selective MFI-type pores, and the resulting hydrogen species diffused through the Brønsted acid sites and was then captured by adsorption-activated CO₂ on the surface of CuZnZrO_x. Hydrogen removal during this accelerated dehydroaromatization promoted the generation of aromatics (40, 57). When the removal of hydrogen was impeded, the aromatic precursor olefins were hydrogenated to form the corresponding alkanes or were further hydrogenolyzed to CH₄ or C₂H₆.

In addition, depolymerized liquid fragments can act as solvents (e.g., *n*-hexadecane; fig. S46) thereby alleviating the heat and mass transfer issues in the solvent-free depolymerization of polymers and providing a starting point for the further optimization and scaling up of the process. We present the route for the coupled and efficient conversion of hydrogen-rich (PE) and hydrogen-deficient (CO₂) feedstocks to produce high-value chemical feedstocks. In the current developed HZSM-5 + CuZnZrO_x system, incorporation of carbon atoms from CO₂ into the aromatic species was not observed. However, in examples of the co-conversion of short-chain alkanes (C₄ to C₆) with CO₂ on a fixed bed, a small amount of CO₂ has been found to enter the aromatic ring via a ketene intermediate (58, 59). In the context of coupling the upgrading of PE and CO₂, the incorporation of CO₂ into aromatics may further increase the yield of aromatics and promote CO₂ conversion. This route is crucial to provide stronger environmental and economic incentives for the global management of plastic and CO₂ waste resources.

We demonstrated an aromatization–hydrogen capture process for converting of PE and CO₂ into valuable aromatics and CO over a multifunctional ZEO-OX catalyst. According to the catalytic and characterization results, we proposed a microscopic mechanism that elucidated the pathway of hydrogen capture mediated by CO₂. In this mechanism, the hydrogen species resulting from PE aromatization at the HZSM-5 acid sites migrated to the neighboring CuZnZrO_x hydrogen acceptor pool for carbon dioxide reduction. This prevented secondary reactions such as hydrogenation and hydrogenolysis of the aromatic precursor olefins. The highest aromatic yield (62.5 wt %) was achieved at mild conditions (380°C), with a CO₂ conversion amount up to 1.5 mmol g_{PE}^{−1} (20 bar CO₂). The coupled mechanism of aromatization and hydrogen-capture provides new ideas for the conversion and recycling of hydrogen-rich polyolefins and hydrogen-deficient CO₂ waste resources.

MATERIALS AND METHODS

Synthesis of zeolite

Na-ZSM-5 was synthesized with tetraethyl orthosilicate (TEOS; >99%, Macklin), aluminum nitrate nonahydrate [Al(NO₃)₃·9H₂O, analytical reagent, Sinopharm Chemical Reagent Co. Ltd.], tetrapropylammonium hydroxide (TPAOH; 25 wt %, Macklin), sodium hydroxide (NaOH; 99%, Innoschem Science & Technology Co. Ltd.), urea [CO(NH₂)₂, AR, Sinopharm Chemical Reagent Co. Ltd.], and water with molar compositions of 1.0 SiO₂, *x*Al₂O₃, 0.3 TPAOH, 0.15 NaOH, 1.2 urea, and 50 H₂O (*x* = 0.0031, 0.0062, and 0.02). The gel was prepared by vigorously stirring the synthesis materials at room temperature for 12 hours. The crystallization was carried out at 180°C for 48 hours under static conditions. The colloidal solid were separated on a centrifuge and washed with water. The solid products were dried at 120°C for 12 hours and calcined at 550°C (the ramping rate of 5°C min^{−1}) for 5 hours.

H-ZSM-5 was synthesized by amine exchange. Na-ZSM-5 zeolite was dispersed within ammonium chloride (NH₄Cl, AR, Aladdin) aqueous solution (1 M; the mass ratio of liquid to solid is 30) and treated at 80°C for 2 hours and repeated three times. The solid were separated on a centrifuge, washed with water, dried at 120°C for 12 hours, and calcined at 550°C (the ramping rate of 5°C min^{−1}) for 5 hours. Moreover, H-Y (Si/Al = 30), H-Beta (Si/Al = 25), and H-MOR (Si/Al = 25) zeolites were purchased from Nankai University Catalyst Co.

S-1 was synthesized with TEOS, TPAOH, NaOH, CO(NH₂)₂, and water with molar compositions of 1.0 SiO₂, 0.3 TPAOH, 0.15 NaOH, 1.2 urea, and 50 H₂O. The gel was prepared by vigorously stirring the synthesis materials at room temperature for 12 hours. The mixture was thermally treated at 180°C for 48 hours for the crystallization. The colloidal solid were separated on a centrifuge and washed with water. The solid products were dried at 120°C for 12 hours and calcined at 550°C (the ramping rate of 5°C min^{−1}) for 5 hours. After calcination, the same amine exchange process was implemented as well as Na-ZSM-5.

Metal oxide catalysts

M-zinc-zirconium oxide (MZnZrO_x, M = Cu, Co, Fe, Ni) were synthesized by the one-pot sol-gel method and subsequent calcination. The molar ratio of M:Zn:Zr is 5:2:2. Here, CuZnZrO_x was synthesized with copper nitrate trihydrate [Cu(NO₃)₃·3H₂O, AR, Sinopharm Chemical Reagent Co. Ltd.], zinc acetate [Zn(CH₃COO)₂, 99.5% Innoschem Science & Technology Co. Ltd.], zirconium nitrate pentahydrate [Zr(NO₃)₄·5H₂O, 98%, Innoschem Science & Technology Co. Ltd.], citric acid monohydrate (AC; 99.8%, Aladdin), ethylene glycol (EG, 99.9%, Aladdin), and water with molar compositions of 1.0 M (total of Cu²⁺, Zn²⁺, and Zr⁴⁺), 1.2 AC, 0.24 EG, and 40.0 H₂O. The gel was prepared by vigorously stirring the mixing materials at 80°C for 4 hours. The solid products were dried at 120°C for 12 hours and calcined at 400°C (the ramping rate of 2°C min^{−1}) for 4 hours. CoZnZrO_x, FeZnZrO_x, as well as NiZnZrO_x were also prepared as the same method, which the ferric nitrate nonahydrate hexahydrate [Fe(NO₃)₃·9H₂O, AR], nickel nitrate hexahydrate [Ni(NO₃)₃·6H₂O, AR], and cobaltous nitrate [Co(NO₃)₂·6H₂O, AR] were used that purchased from Sinopharm Chemical Reagent Co. Ltd.

ZEO-OX catalyst

The ZEO-OX composite catalyst was prepared by grinding HZSM-5 (0.2 g) and CuZnZrO_x (0.1 g) oxide powders in a mortar (powder

mortar grind). For comparison, other physically mixed catalysts (HZSM-5 + MZnZrO_x) were also prepared in a same manner.

Catalyst characterization

XRD patterns of metal oxides and zeolites were obtained on a Rigaku SmartlabSE x-ray diffractometer equipped with $\text{CuK}\alpha$ radiation source ($\lambda = 1.5418 \text{ \AA}$), operated at 40 mA and 40 kV. N_2 physisorption at -196°C was obtained on a Micromeritics ASAP-2460 gas adsorption system.

Before the test, the samples were treated in a dynamic vacuum ($<100 \text{ Pa}$) at 300°C for 3 hours. Analysis used Brunauer-Emmett-Teller surface area and Barrett-Joyner-Halenda adsorption mesopore size. TEM, high-angle annular dark-field TEM and EDX elemental mapping were obtained on Tecnai F20 electron microscope operated at an electron acceleration voltage of 200 kV. SEM and EDS spectroscopy layered elemental mappings were obtained on ZEISS Gemini 300.

NH_3 -TPD was obtained on a Micromeritics TP-5080 instrument equipped with a thermal conductivity detector (TCD) detector. Before the test, the sample (100 mg, 45 to 60 mesh) was treated at 550°C in air for 2 hours and then in N_2 for 1 hour. A 10% NH_3/N_2 gas mixture (30 ml min^{-1}) was dosed to sample at 100°C for 1 hour and switched to pure N_2 flow (30 ml min^{-1}) until the baseline was smooth at the same temperature. The profiles were collected at 100° to 600°C at the ramping of $10^\circ\text{C min}^{-1}$. The adsorbed amount of NH_3 was determined in the temperature range of 100° to 500°C of the profile.

CO_2 -TPD was obtained on a Micromeritics TP-5080 instrument. Before the test, the sample (50 mg) was treated at 300°C in a 5% H_2/N_2 (30 ml min^{-1}) for 3 hours. Then, a pure CO_2 gas (30 ml min^{-1}) was dosed to sample at 30°C for 1 hour and switched to N_2 flow (30 ml min^{-1}) until a stable baseline was obtained at the same temperature. The profiles were collected at 30° to 600°C at the ramping of $10^\circ\text{C min}^{-1}$. The adsorbed amount of CO_2 was determined in the temperature range of 30° to 450°C of the profile.

H_2 -TPD was obtained on a CATLAB (Hidden Analytical Ltd.) instrument equipped with quadrupole mass spectrometer (QIC20). Before the test, the sample (50 mg) was treated at 300°C in a 5% H_2/N_2 (30 ml min^{-1}) for 3 hours, and a pure H_2 gas (30 ml min^{-1}) was dosed to sample at 30°C for 1 hour. Then, the gas was switched to N_2 flow (30 ml min^{-1}) until the baseline was smooth at the same temperature. The profiles were collected at 30° to 500°C at the ramping of $10^\circ\text{C min}^{-1}$.

H_2 -TPR was obtained on a Micromeritics TP-5080 instrument. The profiles were collected at 30° to 500°C at the ramping of $10^\circ\text{C min}^{-1}$ under a 5% H_2/N_2 (30 ml min^{-1}) flow. To make the reduction peaks comparable, the mass of CuZnZrO_x was kept the same (13 mg) for different samples.

Isotope-tracer experiment for CO_2

The isotope labeled experiment was carried out by using $^{13}\text{CO}_2$ gas [no. S16168, enrichment, >99 atomic % ^{13}C , Maotogang Equipments (Shanghai) Co. Ltd.] instead of pure $^{12}\text{CO}_2$ gas with the same reaction. The volatile gas products were analyzed by TCD-mass spectrometry (Pfeiffer TCD-MS instrument) equipped with designed parallel connection system. The liquid products were analyzed by same GC-MS instrument (SHIMADZU GMS-QP2020NX, Japan).

Conversion of PE and carbon dioxide

The catalytic experiments were carried out in a custom-made small high-temperature and high-pressure (upper limit: 450°C , 10 MPa) stainless steel reactor (AISI 316L) equipped with a digital pressure gauge and mechanical agitation (straight blade-style stirrer, 300 rpm) (fig. S2A). The total internal volume of the reactor measured by drainage method is about 45 ml. In a typical procedure, HZSM-5 ($\text{Si}:\text{Al} = 160$, 0.2 g) and reduced CuZnZrO_x (300°C , 20% H_2/N_2 , 3 hours, 0.1 g) were thoroughly mixed and loaded into the reactor with an appropriate PE sample (3.0 g) yielding a PE:catalyst ratio of 10. The reactor atmosphere was cycled between CO_2 (20 bar) and atmospheric pressure five times, and then the reactor was filled with CO_2 (5 bar) and sealed. The instrument was equipped with two K-type sheathed thermocouples, one in the electric heating (furnace) and one in the reactor (contact solid/liquid phase). To bring the reactor to the required specified temperature as quickly as possible, the electric heating furnace was preheated to 500°C . When the potholder (quartz cotton wrapped in high temperature aluminum foil) was opened and inserted into the autoclave, the temperature decreased by about 40°C . The heating curve of the whole process is shown in fig. S2B, reaching the required temperature in about 35 min. Because there is a strong endothermic region around 330°C , it indicates that a reaction had begun. Therefore, when the reactor was inserted in the electric heating furnace, the reaction begins to time, i.e., the reported reaction time includes the time required to heat the reactor from room temperature (about 40 min). During the reaction time, the internal temperature of the reactor fluctuates $\pm 3^\circ\text{C}$.

Conversion of model compound and carbon dioxide

The model compounds used include chain aliphatic hydrocarbons [*n*-hexatriacontane (98%, Psaitong) and *n*-octadecene ($>90\%$, Aladdin)], MS-detected *n*-hexene (99%, Innochem Science & Technology Co. Ltd.), and MS-detected cycloaliphatic hydrocarbons [methyl cyclopentane (99%, Macklin), cyclohexane (99.9%, Adamas), and methylcyclohexane (99%, Innochem Science & Technology Co. Ltd.)]. The catalytic reactions were carried out in the same reactor. The 3.0-g model compound was thoroughly mixed with HZSM-5 + CuZnZrO_x and filled with 20-bar CO_2 .

Catalyst recycling and regeneration

For catalyst recovery cycle test, two parallel reactions were performed at 0.3 g HZSM-5 + CuZnZrO_x , 3.0 g LDPE ($M_w = 1.46 \times 10^5 \text{ g mol}^{-1}$, $\bar{D} = 3.40$), 380°C , and 5 bar CO_2 (reaction time, 200 min). The catalyst was recovered from the two reactions, and after drying at 80°C , about 0.55 g of the reaction catalyst was obtained (denoted as catalyst-0). Under the same reaction conditions, 0.3 g of catalyst-0 was taken for a cycle test. When the catalytic dose was insufficient in the next cycle, catalyst-0 was added. For the catalyst regeneration experiment, the catalyst of the third cycle was calcined in the muffle furnace at 550°C for 4 hours to remove organic residues and carbon deposits. The calcined catalyst was reduced in flowing H_2 (20% H_2/N_2 , 60 ml min^{-1}) at 300°C for 3 hours. The performance of the regenerated catalyst recovered by three cycles was evaluated under the same conditions.

Supplementary Materials

This PDF file includes:
Supplementary Methods
Figs. S1 to S46

Tables S1 to 13
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